# THE ASPECT RATIO OF THE TWIN DRAGON IS $1/\varphi$

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Abstract. We show that the geometric aspect ratio of the Twin Dragon — defined via the Gaussian integer $1+i$ — equals $1/\varphi$, where $\varphi = (1+\sqrt{5})/2$ is the golden ratio, and that its major axis makes angle $-\arctan\varphi$ with the horizontal. More generally, for every equal-weight IFS $\{f_k(z) = a(z+t_k)\}$ with centred translations the aspect ratio depends only on $a$ and a single anisotropy parameter $\kappa = |\sum t_k^2|/\sum |t_k|^2 \in [0,1]$. When $a = 1/\lambda$ for a Gaussian integer $\lambda$, the aspect ratio lies in a quadratic field determined by $\mathbf{N}(\lambda^2-1)$. Every metallic ratio $\mu_m = (m+\sqrt{m^2+4})/2$ $(m \geq 1)$ arises as the reciprocal aspect ratio of a plane-filling tile over $\mathbb{Z}[i]$; moreover, $\mathbb{Z}[i]$ is the unique imaginary quadratic ring where collinear digits can produce metallic-ratio aspect ratios.

## 1. Introduction

The Twin Dragon is a classical plane-filling fractal [4] governed by the Gaussian integer $1+i$: its two defining contractions share the linear part $(1-i)/2$, and their translations are symmetric about the origin. Despite this simple, purely arithmetic construction, the attractor has a non-trivial shape that is not immediately obvious from the definition. Visually the Twin Dragon looks somewhat elongated — neither a square nor a thin sliver — which raises the question of its precise aspect ratio.

A natural quantitative measure of shape for a planar set is the *geometric aspect ratio*: the ratio of standard deviations along the principal axes of its area distribution, or equivalently the square root of the ratio of the eigenvalues of the second-moment (covariance) matrix $M$ of the uniform measure on the attractor. This definition gives the expected answers for elementary shapes: an ellipse with semi-axes $\alpha \geq \beta > 0$ has aspect ratio $\beta/\alpha$ (eigenvalues $\alpha^2/4$ and $\beta^2/4$, so the definition directly recovers the ratio of the semi-axes); a $p \times q$ rectangle has aspect ratio $p/q$ (eigenvalues $p^2/12$ and $q^2/12$); and a square has aspect ratio 1.

The approach of characterising IFS attractors through moments of their invariant measures was introduced by Vrscay and Roehrig [10] and systematically developed in [11]. The key observation is that the self-similar measure satisfies a linear fixed-point equation in its moments, which can be solved exactly. When the open set condition holds, Hutchinson's theorem [6] guarantees that the self-similar measure coincides with the normalised Hausdorff measure $\mathcal{H}^s|_A$, so the covariance matrix captures the actual geometry of the attractor at whatever dimension $s$ it carries. In the special case where the similarity dimension $s$ equals the ambient dimension, $\mathcal{H}^s$ reduces to Lebesgue area.

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We apply this method to the Twin Dragon. The covariance matrix turns out to be $M = \frac{1}{5}\begin{pmatrix} 2 & -1 \\ -1 & 3 \end{pmatrix}$, whose eigenvalues involve $\sqrt{5}$, leading to the aspect ratio $1/\varphi$, where $\varphi = (1+\sqrt{5})/2$ is the golden ratio. The appearance of $\varphi$ in a fractal with no pentagonal geometry is unexpected. Although the method of moments for IFS goes back to [10], the closed-form aspect ratio formula for equal-weight IFS with centred translations (Theorem 1 below) and the characterisation of its quadratic field for Gaussian-integer parameters (Corollary 2) are, to our knowledge, new.

*Discovery.* The result was first found empirically using the IFStile software [9], which implements the moment invariants of [10, 11] as part of its algebraic IFS search engine. Among the similarity invariants reported by IFStile for the Twin Dragon, the aspect ratio $1/\varphi$ appeared as a numerical value; the present paper supplies the exact proof and its generalisation to all equal-weight IFS $\{f_k(z) = a(z+t_k)\}$ with centred translations. This is an instance of *experimental mathematics* [3]: a computer experiment suggests a precise conjecture, which is then confirmed by a rigorous proof.

The paper is organised as follows. Section 2 establishes a closed-form aspect ratio formula (Theorem 1) for equal-weight IFS $\{f_k(z) = a(z + t_k)\}$ with centred translations and classifies the quadratic field of the aspect ratio (Corollary 2). Section 3 applies the formula to a two-parameter family of plane-filling tiles parameterised by $x^2-\ell x+N$ (Proposition 4). Section 4 specialises to tiles over $\mathbb{Z}[i]$, proves that every metallic ratio $\mu_m$ arises as the reciprocal aspect ratio of such a tile (Theorem 7), and shows that $\mathbb{Z}[i]$ is the unique imaginary quadratic ring where collinear tiles achieve metallic-ratio aspect ratios (Corollary 9). Section 5 applies the formula to the Twin Dragon and proves the main result.

## 2. The aspect ratio formula

An *iterated function system* (IFS) is a finite collection of contractions $f_1, \dots, f_N$ on a complete metric space. By Hutchinson's theorem [6] there exists a unique non-empty compact set $A$, the *attractor*, satisfying $A = f_1(A) \cup \cdots \cup f_N(A)$. For background see [5]; for self-affine digit tiles in particular, see [7, 8].

Consider the equal-weight IFS $\{f_k(z) = a(z + t_k)\}_{k=1}^N$ where $a \in \mathbb{C}$, $|a| < 1$, and $t_1, \dots, t_N \in \mathbb{C}$ satisfy $\sum_{k=1}^N t_k = 0$. The condition $\sum t_k = 0$ is no loss of generality: shifting all translations by a constant merely translates the attractor without changing the covariance matrix. Define the *digit variance* $\sigma^2 := \frac{1}{N}\sum_k |t_k|^2 > 0$, the *algebraic variance* $\tau := \frac{1}{N}\sum_k t_k^2 \in \mathbb{C}$, and the *digit anisotropy*

$$\kappa := \frac{|\tau|}{\sigma^2} = \frac{|\sum t_k^2|}{\sum |t_k|^2} \in [0, 1].$$

The bound $\kappa \le 1$ follows from the triangle inequality $|\sum t_k^2| \le \sum |t_k^2| = \sum |t_k|^2$. When all $t_k$ are collinear ($t_k = e^{i\theta} s_k$ with $s_k \in \mathbb{R}$), one has $\kappa = 1$; vertices of a regular $N$-gon ($N \ge 3$) give $\kappa = 0$. The Twin Dragon is the case $N = 2$, $a = (1 - i)/2$, $t_k = \pm 1$ ($\sigma^2 = 1$, $\kappa = 1$), studied in Section 5.

**Theorem 1.** *Let $N \ge 2$, let $t_1, \dots, t_N \in \mathbb{C}$ satisfy $\sum_{k=1}^N t_k = 0$ and $\sigma^2 := \frac{1}{N}\sum_{k=1}^N |t_k|^2 > 0$, let $\kappa = |\sum t_k^2|/\sum |t_k|^2$, and let $a \in \mathbb{C}$ with $|a| < 1$. The*

*invariant equal-weight measure of the IFS* $\{f_k(z) = a(z + t_k)\}_{k=1}^N$ *has aspect ratio*

$$\mathrm{AR}^2 = \frac{|1 - a^2| - (1 - |a|^2)\,\kappa}{|1 - a^2| + (1 - |a|^2)\,\kappa}. \tag{1}$$

*In particular:*

(a) AR *depends on the translations only through* $\kappa$.
(b) *For fixed* $a$, AR *is strictly decreasing in* $\kappa$: $\kappa = 0$ (*isotropic digit set*) *gives* $\mathrm{AR} = 1$, *and* $\kappa = 1$ (*collinear translations*) *gives the most elongated attractor.*
(c) $\mathrm{AR} = 0$ *if and only if* $a \in \mathbb{R}$ *and* $\kappa = 1$.

*Proof.* Let $Z$ be distributed according to the invariant measure $\mu$. The fixed-point equation

$$E[h(Z)] = \frac{1}{N}\sum_{k=1}^{N} E\big[h\big(a(Z + t_k)\big)\big] \tag{2}$$

applied to $h(z) = z$ gives $E[Z] = aE[Z]$, hence $E[Z] = 0$. Applying (2) to $h(z) = z^2$:

$$E[Z^2] = \frac{a^2}{N}\sum_{k=1}^{N}(E[Z^2] + t_k^2) = a^2E[Z^2] + a^2\tau,$$

so $E[Z^2] = a^2\tau/(1 - a^2)$. Applying (2) to $h(z) = |z|^2$:

$$E[|Z|^2] = \frac{|a|^2}{N}\sum_{k=1}^{N}(E[|Z|^2] + |t_k|^2) = |a|^2E[|Z|^2] + |a|^2\sigma^2,$$

so $E[|Z|^2] = |a|^2\sigma^2/(1 - |a|^2)$. The ellipticity ratio is

$$\frac{|E[Z^2]|}{E[|Z|^2]} = \frac{|a|^2|\tau|/(|1 - a^2|)}{|a|^2\sigma^2/(1 - |a|^2)} = \frac{(1 - |a|^2)\,\kappa}{|1 - a^2|},$$

which depends on the digit set only through $\kappa$. The standard identity $\mathrm{AR}^2 = (1 - \rho)/(1 + \rho)$ with $\rho = |E[Z^2]|/E[|Z|^2]$ yields (1). Since $\rho \in [0, 1]$ (the numerator never exceeds the denominator: $(1-|a|^2)\kappa \le 1-|a|^2 \le |1-a^2|$, the first by $\kappa \le 1$ and the second by the triangle inequality), we have $\mathrm{AR}^2 \ge 0$. The right-hand side of (1) is manifestly decreasing in $\kappa$ (both $|1 - a^2|$ and $1 - |a|^2$ are positive), and setting $\kappa = 0$ gives $\mathrm{AR} = 1$. Equality $\mathrm{AR} = 0$ (i.e. $\rho = 1$) requires $(1 - |a|^2)\kappa = |1 - a^2|$, which forces both $\kappa = 1$ and $|1 - a^2| = 1 - |a|^2$; the latter holds iff $a^2 \ge 0$, i.e. $a \in \mathbb{R}$. □

**Remark 1.** *The formula extends to sign-flip IFS* $f_k(z) = \varepsilon_k a(z + t_k)$ *with* $\varepsilon_k \in \{+1, -1\}$. *The centering condition becomes* $\sum \varepsilon_k t_k = 0$; *when it holds,* $\varepsilon_k^2 = 1$ *ensures that* $E[Z^2]$ *and* $E[|Z|^2]$ *reduce to the same expressions as in Theorem 1, so* (1) *applies as stated. When the IFS is not centred* ($\sum \varepsilon_k t_k \neq 0$), *the formula still holds with* $\kappa$ *replaced by*

$$\tilde{\kappa} = \frac{|\tau - \beta^2(1 - a^2)|}{\sigma^2 - |\beta|^2(1 - |a|^2)}, \qquad \beta = \frac{\sum \varepsilon_k t_k}{N - a\sum \varepsilon_k}.$$

*For* $N = 2$ *every sign pattern gives* $\tilde{\kappa} = \kappa$ *(signs are invisible); for* $N \ge 3$ *they enlarge the range of achievable aspect ratios.*

When the contraction ratio is the reciprocal of a Gaussian integer and the translations are collinear ($\kappa = 1$), the aspect ratio is a quadratic irrational whose field is determined by a single Gaussian norm.

**Corollary 2.** *Let $\lambda \in \mathbb{Z}[i]$ with $|\lambda| > 1$ and $\lambda \notin \mathbb{R}$, and set $a = 1/\lambda$. For any IFS with $\kappa = 1$, the aspect ratio* (1) *lies in $\mathbb{Q}(\sqrt{d}\,)$, where $d$ is the square-free part of $\mathbf{N}(\lambda^2 - 1)$. Explicitly,*

$$\mathrm{AR}^2 = \frac{\sqrt{\mathbf{N}(\lambda^2-1)} - (\mathbf{N}(\lambda)-1)}{\sqrt{\mathbf{N}(\lambda^2-1)} + (\mathbf{N}(\lambda)-1)}. \tag{3}$$

*Proof.* Substitute $|a|^2 = 1/\mathbf{N}(\lambda)$, $|1-a^2| = \sqrt{\mathbf{N}(\lambda^2-1)}/\mathbf{N}(\lambda)$, and $\kappa = 1$ into (1). □

TABLE 1. Aspect ratios for small Gaussian integers $\lambda$, illustrating Corollary 2. The quadratic field $\mathbb{Q}(\sqrt{d}\,)$ is determined by the square-free part of $\mathbf{N}(\lambda^2 - 1)$.

| $\lambda$ | $\lambda^2 - 1$ | $\mathbf{N}(\lambda^2-1)$ | $d$ | AR |
|---|---|---|---|---|
| $1+i$ | $-1+2i$ | 5 | 5 | $1/\varphi = (\sqrt{5}-1)/2 \approx 0.618$ |
| $2+i$ | $2+4i$ | 20 | 5 | $1/\varphi^3 = \sqrt{5}-2 \approx 0.236$ |
| $1+2i$ | $-4+4i$ | 32 | 2 | $\sqrt{2}-1 \approx 0.414$ |

The first two rows of Table 1 share the same quadratic field $\mathbb{Q}(\sqrt{5})$ because the rational prime 5 divides both norms. This is not a coincidence: the Cayley-type transform (3) maps $\sqrt{\mathbf{N}(\lambda^2-1)}$ to an element of the real quadratic field $\mathbb{Q}(\sqrt{d}\,)$, and the particular metallic ratio that appears is controlled by the arithmetic of this field. In the first row, $\sqrt{5}$ yields a rational expression in $\varphi$ thanks to the identity $\varphi^2 = \varphi + 1$: the golden ratio is the fundamental unit of the ring $\mathbb{Z}[\varphi]$, so every element of $\mathbb{Q}(\sqrt{5})$ in $(0,1)$ of the form $(\sqrt{5}-n)/(\sqrt{5}+n)$ is a power of $1/\varphi$. The third row gives the *silver ratio* $\delta_S = \sqrt{2} - 1$ (fundamental unit of $\mathbb{Z}[\sqrt{2}]$).

Corollary 2 assumes $\kappa = 1$. The following extension covers arbitrary $\kappa$.

**Corollary 3.** *For any IFS $\{f_k(z) = a(z + t_k)\}$ with $a = 1/\lambda$, $\lambda \in \mathbb{Z}[i] \setminus \mathbb{R}$, and digits $t_k \in \mathbb{Z}[i]$, the anisotropy satisfies $\kappa^2 \in \mathbb{Q}$, and $\mathrm{AR}^2$ lies in $\mathbb{Q}(\sqrt{d}\,)$ where $d$ is the square-free part of $\mathbf{N}(\lambda^2-1)\cdot\big(\mathbf{N}(\lambda^2-1) - \kappa^2(\mathbf{N}(\lambda)-1)^2\big)$. In particular, the quadratic field of* AR *depends only on $\lambda$ and $\kappa^2$, not on the individual digits.*

*Proof.* Write $\tau = \sum (t_k - \bar{t})^2 \in \mathbb{Z}[i]$ and $\sigma^2 = \sum |t_k - \bar{t}|^2 \in \mathbb{Z}$. Then $\kappa^2 = |\tau|^2/\sigma^4 = \mathbf{N}(\tau)/\sigma^4$, which is rational since $\mathbf{N}(\tau) \in \mathbb{Z}$. Substituting into (1) gives

$$\mathrm{AR}^2 = \frac{|1-a^2| - (1-|a|^2)\kappa}{|1-a^2| + (1-|a|^2)\kappa} = \frac{\sqrt{R} - (N-1)\kappa}{\sqrt{R} + (N-1)\kappa},$$

where $R = \mathbf{N}(\lambda^2-1)/N^2$. Clearing $N^2$: the numerator and denominator are $\sqrt{\mathbf{N}(\lambda^2-1)} \mp (N-1)\kappa$. Rationalising, $\mathrm{AR}^2 = 1 - 2(N-1)\kappa/\big(\sqrt{\mathbf{N}(\lambda^2-1)} + (N-1)\kappa\big)$, and since $(N-1)\kappa$ is $\sqrt{\mathbf{N}(\tau)}/\sigma^2 \cdot (N-1)/N$ — a rational multiple of $\sqrt{\mathbf{N}(\tau)}$, while $\sqrt{\mathbf{N}(\lambda^2-1)}$ is algebraic of degree $\leq 2$, $\mathrm{AR}^2$ lies in a quadratic extension of $\mathbb{Q}$. □

**Remark 2.** *For collinear translations ($\kappa = 1$), formula* (1) *shows that* AR *depends on $a$ only through $(|a|^2, |1-a^2|)$: the attractor is nearly round when $|1-a^2|$ is large relative to $1-|a|^2$, and degenerates to a segment when $|1-a^2| \approx 1-|a|^2$ (i.e. $a$ nearly real). For every $a$ in the family, $\mathrm{AR}(a) \leq |a|$, with equality if and only if $a$ is purely imaginary. Indeed, $|1-a^2| \leq 1 + |a|^2$ (triangle inequality), with equality iff*

$a^2/|a^2| = -1$, *i.e.* $a \in i\mathbb{R}$. *Substituting* $|1-a^2| = 1+|a|^2$ *into* (1) *gives* $\mathrm{AR}^2 = |a|^2$. *Figure 1 shows* AR *over the entire sub-family* $|a| = 1/\sqrt{2}$.

**Remark 3.** *The covariance matrix also determines the* orientation *of the principal axes. Writing* $E[Z^2] = \rho e^{2i\psi}$ $(\rho > 0)$ *for* $Z = X + iY$ *with* $E[Z] = 0$, *the matrix decomposes as*

$$M = \frac{E[|Z|^2]}{2}\, I + \frac{\rho}{2}\begin{pmatrix} \cos 2\psi & \sin 2\psi \\ \sin 2\psi & -\cos 2\psi \end{pmatrix},$$

*whose larger eigenvalue has eigenvector* $(\cos\psi, \sin\psi)$. *Hence the major axis of the attractor makes angle*

$$\psi = \tfrac{1}{2}\arg\!\left(\tfrac{a^2\tau}{1-a^2}\right) = \tfrac{1}{2}\arg\!\left(\tfrac{a^2}{1-a^2}\right) + \tfrac{1}{2}\arg(\tau)$$

*with the positive* $x$*-axis. The first term depends only on the contraction* $a$*; the second rotates the principal axes by half the argument of the algebraic variance* $\tau = \frac{1}{N}\sum t_k^2$. *For collinear translations* $\tau > 0$ *and the second term vanishes.*

## 3. The $\lambda_{N,\ell}$ family

A natural two-parameter family of plane-filling tiles arises by applying Theorem 1 to collinear digits. For integers $N \geq 2$ and $0 \leq \ell < 2\sqrt{N}$, let $\lambda_{N,\ell}$ be a root of $x^2 - \ell x + N$ (so $|\lambda_{N,\ell}|^2 = N$ and $\lambda_{N,\ell} \notin \mathbb{R}$). The digits $\{0, 1, \ldots, N-1\}$ form a complete residue system (CRS) of $\mathbb{Z}[\lambda_{N,\ell}]$ modulo $\lambda_{N,\ell}$. Indeed, every element $a + b\lambda_{N,\ell} \in \mathbb{Z}[\lambda_{N,\ell}]$ satisfies $\lambda_{N,\ell}(a + b\lambda_{N,\ell}) = a\lambda_{N,\ell} + b(\ell\lambda_{N,\ell} - N)$ (using $\lambda_{N,\ell}^2 = \ell\lambda_{N,\ell} - N$), whose integer part is $-bN$; hence $a + b\lambda_{N,\ell} \equiv 0 \pmod{\lambda_{N,\ell}}$ forces $b = 0$ and $N \mid a$, so $\lambda_{N,\ell}\mathbb{Z}[\lambda_{N,\ell}] \cap \mathbb{Z} = N\mathbb{Z}$ and the $N$ residues $\{0, \ldots, N-1\}$ are distinct. The $N$-map IFS with $a = 1/\lambda_{N,\ell}$ and digits centred to $\sum t_j = 0$ therefore produces a tile. The aspect ratio and orientation of these tiles admit explicit formulas.

**Proposition 4.** *Let* $N \geq 2$ *and* $0 \leq \ell < 2\sqrt{N}$, *and let* $\lambda = \lambda_{N,\ell}$ *be a root of* $x^2 - \ell x + N$. *For the* $N$*-map tiling IFS with contraction* $a = 1/\lambda$*:*

(i) $|\lambda^2 - 1|^2 = (N+1)^2 - \ell^2$, *and the aspect ratio is*

$$\mathrm{AR}_{N,\ell} = \sqrt{\frac{\sqrt{(N+1)^2 - \ell^2} - (N-1)}{\sqrt{(N+1)^2 - \ell^2} + (N-1)}}\,. \tag{4}$$

(ii) *The major axis of inertia makes angle*

$$\psi_{N,\ell} = -\tfrac{1}{2}\arg(\lambda^2 - 1) \tag{5}$$

*with the positive* $x$*-axis, where* $\lambda^2 - 1 = \frac{1}{2}\big((\ell^2 - 2N - 2) + i\ell\sqrt{4N - \ell^2}\big)$.

(iii) *When* $\ell = 2$ *(the sub-family* $\lambda_N = 1 + i\sqrt{N-1}$*),* (4) *simplifies to* $\mathrm{AR}_N = (\sqrt{N+3} - \sqrt{N-1})/2$, *and* $\tan\psi$ *satisfies* $t^2 + t\sqrt{N-1} - 1 = 0$. *When additionally* $N - 1 = m^2$, *this is the* $m$*-th metallic-ratio equation:* $m = 1$ *gives* $\varphi$, $m = 2$ *gives* $1 + \sqrt{2}$.

*Proof.* Part (i): Expanding $\lambda = (\ell + i\sqrt{4N - \ell^2})/2$ gives

$$|\lambda^2 - 1|^2 = \tfrac{1}{4}\big[(\ell^2 - 2N - 2)^2 + \ell^2(4N - \ell^2)\big] = (N+1)^2 - \ell^2.$$

Substituting $|a|^2 = 1/N$ and $|1 - a^2| = \sqrt{(N+1)^2 - \ell^2}/N$ into (1) yields (4). Part (ii) follows from $a^2/(1-a^2) = 1/(\lambda^2 - 1)$ and Remark 3. Part (iii): Setting $\ell = 2$ gives $(N+1)^2 - 4 = (N-1)(N+3)$; the AR formula rationalises to $(\sqrt{N+3} -$

$\sqrt{N-1})/2$. For the tangent, $\lambda^2-1=(1-N)+2i\sqrt{N-1}$; the double-angle identity $\tan 2\psi = 2\tan\psi/(1-\tan^2\psi) = 2/\sqrt{N-1}$ yields the stated equation. □

Selected values:

| $N$ | $\ell$ | $\lambda_{N,\ell}$ | field | $\mathrm{AR}_{N,\ell}$ |
|---|---|---|---|---|
| 2 | 0 | $i\sqrt{2}$ | $\mathbb{Q}$ | $1/\sqrt{2}\approx 0.707$ (rectangle) |
| 2 | 1 | $(1+i\sqrt{7})/2$ | $\mathbb{Q}(\sqrt{2})$ | $\approx 0.691$ (tame twindragon) |
| 2 | 2 | $1+i$ | $\mathbb{Q}(\sqrt{5})$ | $1/\varphi\approx 0.618$ |
| 5 | 2 | $1+2i$ | $\mathbb{Q}(\sqrt{2})$ | $\sqrt{2}-1\approx 0.414$ |
| 5 | 4 | $2+i$ | $\mathbb{Q}(\sqrt{5})$ | $1/\varphi^3\approx 0.236$ |

The "field" column records $\mathbb{Q}(\sqrt{d}\,)$ where $d$ is the square-free part of $(N+1)^2-\ell^2$. The entry $(N,\ell)=(2,1)$ recovers the tame twindragon (see below), confirming that it belongs to the same two-parameter family. The entry $(N,\ell)=(5,4)$ gives $\lambda_{5,4}=2+i$ — the same contraction as the 2-map tile in row 2 of Table 1, illustrating again that AR depends only on $a=1/\lambda$ (Theorem 1).

For fixed $\ell$, $\mathrm{AR}_{N,\ell}\to 0$ as $N\to\infty$ (stronger contraction elongates the attractor). For fixed $N$, increasing $\ell$ toward $2\sqrt{N}$ drives $a$ toward the real axis and $\mathrm{AR}\to 0$; setting $\ell=0$ gives $\lambda=i\sqrt{N}$ (purely imaginary $a$) and $\mathrm{AR}=1/\sqrt{N}$, the maximum for a given $N$. For $N=2$ the IFS $\{(i/\sqrt{2})(z\pm 1)\}$ has attractor the rectangle $[-1,1]\times[-\sqrt{2},\sqrt{2}]$ (one verifies that $f_1$ and $f_2$ map it to the top and bottom halves); this rectangle tiles the plane by translation.

*Tame twindragon.* The entry $(N,\ell)=(2,1)$ gives $\lambda=(1+i\sqrt{7})/2$ and the attractor is the *tame twindragon* [1, p. 550], the attractor of $\{f_{1,2}(z)=a(z\pm 1)\}$ with $a=(1-i\sqrt{7})/4$. One computes $1-a^2=(11+i\sqrt{7})/8$, hence $|1-a^2|=\sqrt{2}$, and Theorem 1 gives

$$\mathrm{AR}^2=\frac{\sqrt{2}-\frac{1}{2}}{\sqrt{2}+\frac{1}{2}}=\frac{2\sqrt{2}-1}{2\sqrt{2}+1}=\frac{9-4\sqrt{2}}{7}\approx 0.478,\qquad \mathrm{AR}\approx 0.691.$$

This tile arises as the numeration digit set for the complex base $\beta=(1+i\sqrt{7})/2$ (satisfying $\beta^2-\beta+2=0$ and $|\beta|^2=2$) via $a=1/\beta$;[1] it is marked in Figure 1.

The connection to metallic ratios in part (iii) is not accidental. The $m$-th metallic ratio $\mu_m=(m+\sqrt{m^2+4})/2$ is the fundamental unit of $\mathbb{Z}[\mu_m]\subset\mathbb{Q}(\sqrt{m^2+4})$. In the $\ell=2$ sub-family, $\sqrt{m^2+4}=\sqrt{N+3}$ and the Cayley-type transform (3) maps $(N+1)^2-4=(N-1)(N+3)$ into $\mathbb{Q}(\sqrt{N+3})$. For $N=2$ and $N=5$ the aspect ratio is an exact power of the fundamental unit ($1/\varphi$ and $\sqrt{2}-1$ respectively); for other $N$ it remains in $\mathbb{Q}(\sqrt{N+3})$ but is not generally a unit.

**Remark 4.** *Both the aspect ratio and the principal-axis orientation of the $\lambda_{N,\ell}$-family are governed by a single element of the ring of integers: $\lambda^2-1=\ell\lambda-(N+1)\in\mathbb{Z}[\lambda]$. Its norm $(N+1)^2-\ell^2$ determines* AR *via* (4)*, and its argument determines $\psi$ via* (5)*. This admits a hierarchical interpretation. The random-iteration representation $Z=\sum_{n\geq 1}a^n\varepsilon_n$ ($\varepsilon_n$ i.i.d. from the centred digits) decomposes $E[Z^2]$ as $\sum_{n\geq 1}\sigma^2a^{2n}$: each level $n$ of the IFS contributes $\sigma^2/N^n$ to the anisotropy along direction $2n\arg(a)=-2n\arg(\lambda)$. The geometric sum of these rotated contributions points in the direction $\arg\big(a^2/(1-a^2)\big)=-\arg(\lambda^2-1)$, which is precisely twice the principal-axis angle. At the boundary $\ell=0$ the vector $\lambda^2-1=-(N+1)$ is real*

[1]The attractor is the set $\{\sum_{k\geq 1}d_k\beta^{-k}:d_k\in\{-1,+1\}\}\subset\mathbb{C}$.

*negative, giving* $\psi = -\pi/2$ *(vertical major axis); as* $\ell \to 2\sqrt{N}$ *the vector tends to* $2(N-1) > 0$*, giving* $\psi \to 0$ *(horizontal, degenerate).*

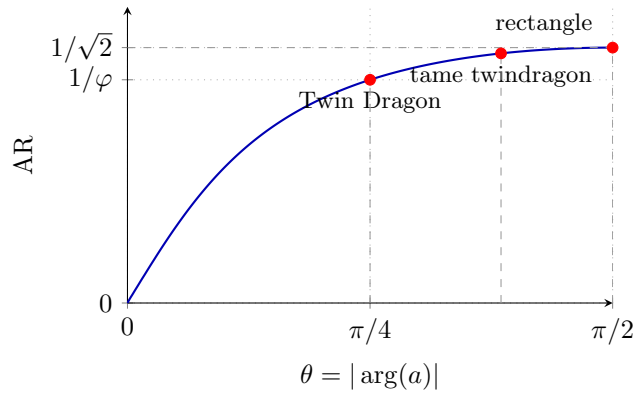


FIGURE 1. Aspect ratio $\mathrm{AR}(\theta)$ of the attractor of $\{a(z \pm 1)\}$ for $|a| = 1/\sqrt{2}$, as a function of $\theta = |\arg(a)|$. The Twin Dragon ($\theta = \pi/4$), the tame twindragon ($\theta = \arctan\sqrt{7}$), and the rectangle attractor ($\theta = \pi/2$) are marked in red.

## 4. Tiles over $\mathbb{Z}[i]$

In this section we specialise to tiles over $\mathbb{Z}[i]$: the contraction is $a = 1/\lambda$ for a Gaussian integer $\lambda$ and the translations form a CRS modulo $\lambda$. Theorem 1 then applies with $|a|^2 = 1/N$ and $\kappa$ determined by the choice of CRS.

The dependence on $\kappa$ is already visible for a single expansion factor. Take $\lambda = 2 + i$ ($N = 5$). The collinear digit set $\{0, 1, 2, 3, 4\}$ gives $\kappa = 1$ and $\mathrm{AR} = 1/\varphi^3$ (row $(5, 4)$ of the table in Section 3). Replacing it by the two-dimensional CRS $\{0,\ 1,\ -i,\ i,\ 1{+}i\}$ (which is still a complete residue system of $\mathbb{Z}[i]$ modulo $2{+}i$, since $i \equiv 3 \pmod{2+i}$) yields $\sigma^2 = 4/5$, $\tau = (-8 + 6i)/25$, and $\kappa = 1/2$. Substituting into (1):

$$\mathrm{AR}^2 = \frac{\sqrt{5}-1}{\sqrt{5}+1} = \frac{3-\sqrt{5}}{2} = \frac{1}{\varphi^2}, \qquad \mathrm{AR} = \frac{1}{\varphi}.$$

Thus the same golden ratio governs this 5-map tile with $\kappa = 1/2$ as it does the Twin Dragon ($N = 2$, $\kappa = 1$), while collinear digits for the same $\lambda$ give the smaller value $1/\varphi^3$ (Figure 2). A natural question arises: for which $\kappa$ does the aspect ratio equal $1/\mu_m$, the reciprocal of the $m$-th metallic mean?

**Proposition 5.** *Let* $\lambda = p + qi \in \mathbb{Z}[i] \setminus \mathbb{R}$ *with* $|\lambda|^2 = N$*, and let* $m \geq 1$ *be a positive integer. The digit anisotropy required for* $\mathrm{AR} = 1/\mu_m$ *(the reciprocal* $m$*-th metallic mean* $\mu_m = (m + \sqrt{m^2+4})/2$*) is*

$$\kappa_m^* = \frac{m\,|\lambda^2 - 1|}{\sqrt{m^2+4}\,(N-1)}, \tag{6}$$

*and* $\kappa_m^* \leq 1$ *if and only if* $mq \leq N - 1$*. In particular:*

(a) *For* $m = 1$ *(golden ratio),* $\kappa_1^* \leq 1$ *for every* $\lambda$*, with equality only for* $\lambda = 1 + i$*.*

(b) *The collinear case $\kappa = 1$ corresponds to $m = (N-1)/q$, provided this is an integer; thus the collinear aspect ratio of Corollary 2 equals $1/\mu_{(N-1)/q}$.*
(c) *For $q = 1$ (i.e. $\lambda = p + i$), every metallic ratio $\mu_1, \ldots, \mu_{p^2}$ is accessible.*

*Proof.* Setting $\mathrm{AR}^2 = 1/\mu_m^2$ in (1) and solving for $\kappa$ gives $\kappa = |1 - a^2| \cdot m/\big((1 - |a|^2)\sqrt{m^2+4}\big)$; substituting $a = 1/\lambda$, $|a|^2 = 1/N$, $|1 - a^2| = \sqrt{\mathbf{N}(\lambda^2 - 1)}/N$ yields (6).

The condition $\kappa_m^* \le 1$ becomes $m^2|\lambda^2 - 1|^2 \le (m^2+4)(N-1)^2$. Since $|\lambda^2 - 1|^2 = (N-1)^2 + 4q^2$, this simplifies to $4m^2q^2 \le 4(N-1)^2$, i.e. $mq \le N-1$.

*(a)*: $q \le N - 1$ holds since $N = p^2 + q^2 \ge q + 1$ for $p \ge 1$. Equality $q = N-1$ gives $p^2 = 1 - q(q-1)$; for $q = 1$ this yields $p = 1$, $\lambda = 1 + i$; for $q \ge 2$ one has $p^2 < 0$, which is impossible.

*(b)*: $\kappa_m^* = 1$ iff $m^2q^2 = (N-1)^2$, so $m = (N-1)/q$.

*(c)*: $q = 1$ gives $mq = m \le N - 1 = p^2$. □

**Corollary 6.** *Up to associates, $\kappa_1^* = 1/2$ if and only if $\lambda \in \{2 + i,\ 2 + 3i,\ 1 + 4i\}$ ($N = 5, 13, 17$). More generally, $\kappa_1^* = 1/m$ for an integer $m \ge 1$ only for $m = 1$ ($\lambda = 1 + i$) and $m = 2$ (the three primes above).*

*Proof.* Write $\lambda = p+qi$, $N = p^2+q^2$. Setting $\kappa_1^* = 1/m$ in $|\lambda^2-1|^2 = 5(N-1)^2/m^2$ and using $|\lambda^2 - 1|^2 = (N - 1)^2 + 4q^2$ gives

$$4q^2 = (N-1)^2\big(\tfrac{5}{m^2} - 1\big).$$

For $m = 1$ this is $4q^2 = 4(N-1)^2$, so $q = N-1 = 1$, $N = 2$: only $\lambda = 1 + i$. For $m = 2$: $4q^2 = \frac{1}{4}(N-1)^2$, so $N = 4q + 1$, $p^2 = 4q + 1 - q^2$, and the positivity condition $p^2 \ge 1$ becomes $p^2 + (q - 2)^2 = 5$, i.e. a representation of 5 as a sum of two squares. The three solutions with $q \ge 1$, $p \ge 1$ are $(p, q) = (2, 1), (2, 3), (1, 4)$. For $m \ge 3$: $5/m^2 - 1 < 0$, so $4q^2 < 0$ — impossible. □

**Theorem 7.** *For every integer $m \ge 1$:*

(a) *There exists a self-similar plane-filling tile in $\mathbb{R}^2$ whose aspect ratio equals $1/\mu_m$, the reciprocal of the $m$-th metallic mean.*
(b) *Any such tile over $\mathbb{Z}[i]$ requires at least $N_{\min}(m) = \lceil\sqrt{m}\,\rceil^2 + 1$ maps.*
(c) *When $m$ is a perfect square, $N_{\min}(m) = m + 1$, achieved by the collinear CRS for $\lambda = \sqrt{m} + i$.*

*Proof. (a):* Let $\lambda = 1 + mi \in \mathbb{Z}[i]$. Then $N = |\lambda|^2 = m^2 + 1$ and $\mathrm{Im}(\lambda) = m$. The standard digits $\{0, 1, \ldots, m^2\}$ form a CRS modulo $\lambda$ (§3); the induced $N$-map IFS satisfies the open set condition [1], so its attractor is a plane-filling tile. The collinear digit set has $\kappa = 1$, and Proposition 5(b) gives $m = (N-1)/\mathrm{Im}(\lambda) = m^2/m$, hence $\mathrm{AR} = 1/\mu_m$.

*(b):* For any $\lambda = p+qi$ with $mq \le N-1$ (Proposition 5), we have $N = p^2+q^2 \ge q(m-q) + 1 + q^2 = qm + 1$, which is minimised at $q = 1$, giving $N \ge \lceil\sqrt{m}\,\rceil^2 + 1$.

*(c):* When $m = k^2$, the choice $\lambda = k + i$ yields $N = k^2 + 1 = m + 1$ and the collinear CRS has $\kappa = 1$ with $(N-1)/q = m$. □

**Remark 5.** *Not every integer $N \ge 2$ is the norm of a Gaussian integer with non-zero imaginary part. By the sum-of-two-squares theorem, $N = p^2 + q^2$ with $q \ne 0$ if and only if every prime factor of $N$ congruent to $3 \pmod 4$ appears to an even power. The first admissible values are $N = 2, 4, 5, 8, 9, 10, 13, \ldots$; in particular $N = 3, 6, 7, 11, 14, 15$ are not representable.*

**Remark 6.** *Item (b) of Proposition 5 recovers every row of Table 1 as a special case:* $\lambda = 1+i$ *gives* $m = 1$ *and* $\mathrm{AR} = 1/\mu_1 = 1/\varphi$*;* $\lambda = 2+i$ *gives* $m = 4$ *and* $\mathrm{AR} = 1/\mu_4 = 1/(2+\sqrt{5}) = \sqrt{5}-2 = 1/\varphi^3$*;* $\lambda = 1+2i$ *gives* $m = 2$ *and* $\mathrm{AR} = 1/\mu_2 = \sqrt{2}-1$*. Thus the collinear aspect ratios of Section 2 and the golden-ratio tiles of Table 2 are part of a single metallic-ratio spectrum parametrised by* $\kappa$*. Table 2 lists CRS digit sets achieving* $\kappa_1^*$ *(i.e.* $\mathrm{AR} = 1/\varphi$*) for small Gaussian primes* $\lambda$*.*

TABLE 2. Complete residue systems $D \subset \mathbb{Z}[i]$ (modulo $\lambda$, centred to $\sum t_k = 0$) achieving $\mathrm{AR} = 1/\varphi$.

| $\lambda$ | $N$ | $\kappa^*$ | *digit set* $D$ |
|---|---|---|---|
| $1+i$ | 2 | 1 | $\{0, \pm 1\}$ |
| $2+i$ | 5 | $1/2$ | $\{0, 1, -i, i, 1+i\}$ |
| $1+2i$ | 5 | $\sqrt{2/5}$ | $\{-1-2i, -1, i, 1, 1+i\}$ |
| $1+3i$ | 10 | $\sqrt{13/45}$ | $\{-2, -2+2i, -1-i, -1, -1+i, -i, 0, i, 1-2i, 1\}$ |
| $2+3i$ | 13 | $1/2$ | $\{3-2i, 5i, 3-5i, 3, \ldots, 12\}$ |
| $1+4i$ | 17 | $1/2$ | $\{-3, \ldots, 2+2i\}$ *(17 digits)* |

*Rows 3 and 4 have irrational* $\kappa^*$ *(*$\sqrt{2/5}$ *and* $\sqrt{13/45}$*), yet Gaussian-integer digits realise these values exactly. Corollary 6 explains why rows 2, 5, and 6 all share* $\kappa^* = 1/2$*: only three Gaussian primes attain this value, and the proof reduces to the unique representation* $5 = 1^2 + 2^2$*.*

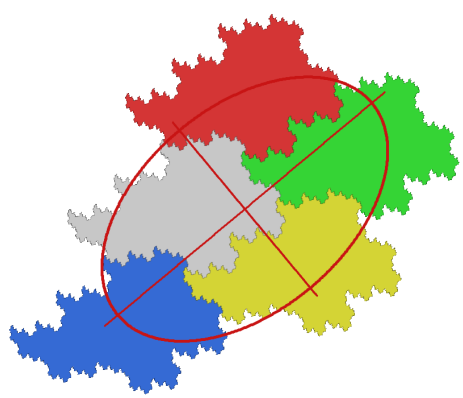

FIGURE 2. The 5-map tile with $\lambda = 2+i$ and digit set $\{0, 1, -i, i, 1+i\}$ ($\kappa = 1/2$). Its 1.5-sigma principal-axis ellipse (red) confirms $\mathrm{AR} = 1/\varphi \approx 0.618$, the same value as the Twin Dragon (Figure 3).

**Proposition 8.** *Let* $\lambda \in \mathbb{C} \setminus \mathbb{R}$ *with* $|\lambda|^2 = N \geq 2$*. Then*

$$|\lambda^2 - 1|^2 = (N-1)^2 + 4\,(\operatorname{Im}\lambda)^2, \tag{7}$$

*and the digit anisotropy* (6) *satisfies* $\kappa_m^* \le 1$ *if and only if* $m \le (N-1)/|\operatorname{Im}\lambda|$. *The collinear case* $\kappa = 1$ *corresponds to* $m = (N-1)/|\operatorname{Im}\lambda|$, *which is a positive integer only when* $|\operatorname{Im}\lambda|$ *divides* $N-1$.

*Proof.* Write $\lambda = x + iy$, so $N = x^2 + y^2$ and $\lambda^2 - 1 = (x^2-y^2-1) + 2ixy$. Then

$$|\lambda^2 - 1|^2 = (x^2-y^2-1)^2 + 4x^2y^2 = (x^2+y^2-1)^2 + 4y^2 = (N-1)^2 + 4y^2,$$

since, by direct expansion, $(x^2-y^2-1)^2 + 4x^2y^2 - (x^2+y^2-1)^2 = 4y^2$. Substituting into (6): $\kappa_m^* \le 1$ iff $m^2\big((N-1)^2 + 4y^2\big) \le (m^2+4)(N-1)^2$, i.e. $4m^2y^2 \le 4(N-1)^2$, so $m \le (N-1)/|y|$. □

**Corollary 9.** *Among all imaginary quadratic integer rings,* $\mathbb{Z}[i]$ *is the unique one where collinear self-similar tiles can have metallic-ratio aspect ratios.*

*Proof.* For $\lambda = p + q\sqrt{-k} \in \mathbb{Z}[\sqrt{-k}]$ ($k \ge 2$, square-free, $q \ne 0$), $\operatorname{Im}\lambda = q\sqrt{k} \notin \mathbb{Q}$, so $(N-1)/\operatorname{Im}\lambda$ is irrational. The same holds for the full ring of integers of $\mathbb{Q}(\sqrt{-3})$: if $\lambda = a + b\omega \in \mathbb{Z}[\omega]$ with $\omega = e^{2\pi i/3}$ and $b \ne 0$, then $\operatorname{Im}\lambda = b\sqrt{3}/2 \notin \mathbb{Q}$. In $\mathbb{Z}[i]$, however, $\operatorname{Im}\lambda = q \in \mathbb{Z}$, and taking $q = 1$ gives $m = (N-1)/1 = p^2$, always a positive integer. □

**Remark 7.** *Proposition 8 shows that the identity* $|\lambda^2-1|^2-(N-1)^2$ *is controlled entirely by* $\operatorname{Im}\lambda$. *Specialising to each ring:* $\mathbb{Z}[i]$*:* $4q^2$*;* $\mathbb{Z}[\omega]$*:* $3b^2$*;* $\mathbb{Z}[\sqrt{-k}]$*:* $4kq^2$. *The upper bound* $m_{\max} = (N-1)/|\operatorname{Im}\lambda|$ *is rational only for* $\mathbb{Z}[i]$. *Nonetheless, by choosing* $\kappa < 1$ *(non-collinear digits),* $\kappa_m^* \le 1$ *is guaranteed for every imaginary quadratic ring whenever* $N$ *is large enough. A computational search over* $\mathbb{Z}[\omega]$ *confirms that complete residue systems achieving* $\kappa = \kappa_1^*$ *(i.e.* $\mathrm{AR} = 1/\varphi$*) exist for all Eisenstein primes of norm* $N \ge 7$. *However, for* $N = 3$ *(*$\lambda = 1-\omega$, $\kappa_1^* = \sqrt{7/20}$*), an exhaustive search over shifts* $|c_k| \le 8$ *fails to achieve* $\kappa_1^*$ *(best approximation* $\approx 3 \times 10^{-5}$*), suggesting that three digits provide too few degrees of freedom in the hexagonal lattice.*

**Remark 8.** *For* collinear *digits (*$\kappa = 1$*) the condition* $m = (N-1)/q$ *(Proposition 5(b)) requires* $p^2 = q(m-q)+1$ *to be a perfect square. Taking* $q = m$ *gives the universal upper bound* $N_{\mathrm{col}}(m) \le m^2+1$, *but smaller* $q$ *may yield a smaller* $N$. *For* $m = k^2$ *a perfect square,* $q = 1$ *gives* $N_{\mathrm{col}}(k^2) = k^2+1 = N_{\min}(k^2)$, *so the collinear and non-collinear minima coincide. In contrast, for non-square* $m$ *a gap opens: e.g.* $N_{\min}(5) = 10$ *while* $N_{\mathrm{col}}(5) = 26$, *and* $N_{\min}(7) = 10$ *while* $N_{\mathrm{col}}(7) = 50$. *In general,* $N_{\mathrm{col}}(m)$ *equals the minimum of* $q^2 + q(m-q) + 1$ *over those* $q \in \{1, \dots, m\}$ *for which* $q(m-q)+1$ *is a perfect square — a Diophantine condition whose behaviour mirrors the distribution of square values of binary quadratic forms.*

## 5. The Twin Dragon

The Twin Dragon $A \subset \mathbb{C}$ is the attractor of the IFS [4]

$$A = f_1(A) \cup f_2(A), \qquad f_{1,2}(z) = \frac{1-i}{2}(z \pm 1). \tag{8}$$

Both maps have the same contraction ratio $r = |(1-i)/2| = 1/\sqrt{2}$, so the similarity dimension is $\alpha = -\log 2/\log r = 2$. The open set condition holds [1, Corollary 3]: the Twin Dragon is a digit tile for $\mathbb{Z}[i]/(1+i)\mathbb{Z}[i]$, so its translates by $\mathbb{Z}[i]$ tile $\mathbb{C}$ with the interiors pairwise disjoint. By [6, §5] the self-similar measure therefore

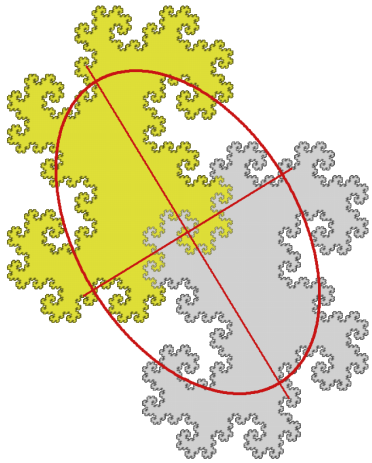

FIGURE 3. The Twin Dragon with its 1.5-sigma principal-axis ellipse (red). The major axis lies along $(1, -\varphi)$ and the minor axis along $(\varphi, 1)$; their ratio of lengths equals the aspect ratio $\mathrm{AR} = 1/\varphi \approx 0.618$.

coincides with $\mathcal{H}^\alpha|_A$. Since $\alpha = 2$ equals the ambient dimension, $\mathcal{H}^2$ is Lebesgue area, and the covariance matrix $M$ describes the actual geometric shape (Figure 3).

Substituting $a = (1-i)/2$ into Theorem 1 (with $N = 2$, $t_k = \pm 1$, so $\sigma^2 = 1$): $|a|^2 = \frac{1}{2}$, $a^2 = -\frac{i}{2}$, $1 - a^2 = \frac{2+i}{2}$, so

$$E[Z^2] = \frac{a^2}{1-a^2} = \frac{-i}{2+i} = \frac{-1-2i}{5}, \qquad E[|Z|^2] = \frac{|a|^2}{1-|a|^2} = 1.$$

The covariance matrix follows via $M_{11} = \frac{1}{2}(E[|Z|^2] + \operatorname{Re} E[Z^2])$, $M_{22} = \frac{1}{2}(E[|Z|^2] - \operatorname{Re} E[Z^2])$, $M_{12} = \frac{1}{2}\operatorname{Im} E[Z^2]$:

$$M = \frac{1}{5}\begin{pmatrix} 2 & -1 \\ -1 & 3 \end{pmatrix}. \tag{9}$$

The matrix $M$ has $\operatorname{tr} M = 1$ and $\det M = \frac{1}{5}$, so its eigenvalues are

$$I_{1,2} = \frac{1 \mp \frac{1}{\sqrt{5}}}{2}. \tag{10}$$

The geometric aspect ratio — the ratio of standard deviations along the principal axes of $M$ — is

$$\sqrt{\frac{I_1}{I_2}} = \sqrt{\frac{\sqrt{5}-1}{\sqrt{5}+1}} = \sqrt{\frac{3-\sqrt{5}}{2}} = \frac{1}{\varphi}, \tag{11}$$

where $\varphi = (1+\sqrt{5})/2$ is the golden ratio.

**Corollary 10.** *The attractor of the IFS* $\{f_{1,2}(z) = \frac{1-i}{2}(z \pm 1)\}$ *has covariance matrix* $M = \frac{1}{5}\left(\begin{smallmatrix} 2 & -1 \\ -1 & 3 \end{smallmatrix}\right)$*, geometric aspect ratio* $1/\varphi = (\sqrt{5}-1)/2 \approx 0.618$*, and major-axis direction* $(1, -\varphi)$ *(angle* $-\arctan\varphi \approx -58.3^\circ$ *with the horizontal), where* $\varphi = (1+\sqrt{5})/2$ *is the golden ratio.*

*Proof.* This is the case $\lambda = 1 + i$ of Corollary 2 ($\mathbf{N}(\lambda^2 - 1) = 5$, $d = 5$), with the orientation computed from the general formula in Remark 3. □

**Remark 9.** *The Twin Dragon is the* $(N, \ell) = (2, 2)$ *instance of Proposition 4, giving* $\tan\psi = -\varphi$ *(i.e.* $\psi \approx -58.3°$*). Thus the same constant* $\varphi$ *governs both the aspect ratio (*$\mathrm{AR} = 1/\varphi$*) and the orientation (*$\tan\psi = -\varphi$*).*

**Remark 10.** *The appearance of* $\varphi$ *is unexpected: the Twin Dragon is defined by the Gaussian integer* $1 + i$*, with no pentagon, decagon, or Fibonacci recurrence in the construction. The number* 5 *— and hence* $\varphi$ *— can be traced to a single Gaussian prime. By the ergodic theorem for random iteration* [2]*,* $\mu$ *is the distribution of the random series* $Z = \sum_{k\geq 1} \varepsilon_k a^k$ *with* $a = (1 - i)/2$ *and* $\varepsilon_k \overset{\text{iid}}{\sim} \{\pm 1\}$*. Since* $1 - a^2 = (2 + i)/2$ *and* $\kappa = 1$ *(real translations), the ellipticity is* $\rho = (1 - |a|^2)\kappa/|1 - a^2| = 1/\sqrt{5}$*, where* $5 = \mathbf{N}(2 + i)$ *is the norm of the Gaussian prime* $2 + i$*. Thus* $\varphi$ *enters via the Gaussian factorisation* $5 = (2 + i)(2 - i)$ *in* $\mathbb{Z}[i]$*, and the identity* $\mathrm{AR}^2 = (1 - \rho)/(1 + \rho)$ *from* (1) *immediately gives* $1/\varphi^2$ *(using* $\varphi^2 = \varphi + 1$*).*

**Further directions.** An explicit closed-form construction of a CRS achieving $\kappa = \kappa^*_m$ for all $N$ would be desirable; this amounts to a system of quadratic Diophantine equations in the digit shifts $c_k \in \mathbb{Z}[i]$ (where $d_k = k + c_k\lambda$), which does not appear to simplify into a universal formula. The sign-flip extension of Remark 1 provides an additional degree of freedom: by choosing $\varepsilon_k \in \{+1, -1\}$ one can tune $\tilde{\kappa}$ away from $\kappa$ and thereby reach aspect ratios inaccessible to the original IFS with uniform signs.

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